

ECO764: Financial Econometrics

Second Assignment: Group

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Total Marks: 70

Instructions

- This is a **group assignment**.
- Each group may consist of **a maximum of five members**. Groups exceeding this limit will not be accepted.
- All questions must be answered using **empirical data and econometric analysis**.
- Students may use statistical software such as **R, Python, Matlab, or Excel**. Students using R may employ packages such as **quantmod, tidyquant, or PerformanceAnalytics** to obtain and analyse financial data.
- Clearly report:
 - Data sources
 - Econometric methodology
 - Model specification
 - Interpretation of empirical results
- All tables, figures, and graphs must be clearly labelled and discussed in the text.
- Include the **code used for estimation** in an appendix.

Submission Guidelines

- Submit a single **PDF report** (maximum 15 pages excluding appendix).

- The report should clearly list the **names and roll numbers of all group members**.
- Attach the dataset and code files used for the empirical analysis.
- File naming format:

`EC0764_Assignment_GroupName.zip`

- The assignment must be submitted via email to:

`wasimad24@gmail.com`

- **Deadline for submission: March 31, Midnight.**

Suggested Data Sources

- NSE India / Yahoo Finance
- MCX Commodity Exchange
- RBI Database on Indian Economy
- NOAA Climate Data
- Indian Meteorological Department
- Baltic Exchange (freight indices)
- European Energy Exchange (carbon prices)

Question 1: Idiosyncratic Volatility and Stock Returns (10 Marks)

Select at least *10 stocks listed on NSE* and collect daily or monthly price data for a period of *not less than 10 years*. Data may be downloaded using the *quantmod* package in R or other financial databases.

Use a broad market index such as *Nifty 50* as the market portfolio.

- (a) Download price data and construct return series for all stocks and the market index. Report descriptive statistics including mean, standard deviation, skewness, and kurtosis. (2 marks)

- (b) Estimate the CAPM regression for each stock:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t}$$

Compute *idiosyncratic volatility (IVOL)* as the standard deviation of the residuals:

$$IVOL_i = \sqrt{Var(\epsilon_{i,t})}$$

(3 marks)

- (c) Sort stocks into *three portfolios (Low, Medium, High IVOL)* based on the estimated idiosyncratic volatility and compute the average returns of each portfolio. (3 marks)
- (d) Estimate the following cross-sectional regression:

$$R_i = \gamma_0 + \gamma_1 IVOL_i + u_i$$

Interpret whether the results support the *idiosyncratic volatility puzzle*. (2 marks)

Question 2: Empirical Analysis of Factor Investing (10 Marks)

This question examines the empirical performance of factor-based investment strategies using open-source financial data.

- (a) Select at least *20 stocks listed on the National Stock Exchange (NSE)* and obtain monthly price data for a sample period of *not less than 10 years*. Data may be downloaded from open sources such as Yahoo Finance using the *quantmod* package in R or other financial databases. Construct monthly return series for all selected stocks. Provide summary statistics for the return series. (2 marks)
- (b) Construct at least *two factor portfolios* based on firm characteristics. Possible factors include:
- Size (market capitalisation)
 - Value (price-to-book or book-to-market ratio)
 - Momentum (past 12-month return)
 - Low-volatility factor

Clearly describe the methodology used to construct the factor portfolios, including the sorting procedure and portfolio formation rules. (3 marks)

- (c) Estimate the following *multi-factor asset pricing model*:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{MKT}(R_{m,t} - R_{f,t}) + \beta_{SMB}SMB_t + \beta_{HML}HML_t + \epsilon_{i,t}$$

Interpret the estimated factor loadings and discuss their economic significance. (3 marks)

- (d) Evaluate the performance of the factor portfolios by computing key risk-adjusted performance measures, including the mean return, standard deviation, and Sharpe ratio. Compare the performance of the factor portfolios with a market benchmark and discuss the implications of your findings. (2 marks)

Question 3: Construction of Continuous Futures Series (10 Marks)

Select one commodity traded on *MCX* (e.g. gold, crude oil, copper).

- (a) Download spot and futures prices for multiple contract maturities. (2 marks)
- (b) Construct a *continuous futures price series* using a rollover method. Plot the resulting series. (3 marks)
- (c) Conduct *unit root tests* (ADF) for both spot and futures prices. (2 marks)
- (d) Test for *cointegration* between spot and futures prices and interpret the economic implications. (3 marks)

Question 4: Price Discovery and Hedging Effectiveness (10 Marks)

Using the spot and futures dataset from Question 2:

- (a) Estimate a *Vector Error Correction Model (VECM)*. Interpret the adjustment coefficients. (3 marks)
- (b) Compute price discovery measures such as:
 - Hasbrouck Information Share

- Component Share
- Garbade–Silber measure

Compare which market leads in price discovery. (4 marks)

(c) Estimate the *optimal hedge ratio*:

$$h^* = \frac{Cov(\Delta S, \Delta F)}{Var(\Delta F)}$$

Evaluate hedging effectiveness using variance reduction. (3 marks)

Question 5: Empirical Identification of Arbitrage Opportunities (10 Marks)

Using the futures dataset from Question 2:

(a) Estimate the theoretical futures price using the cost-of-carry model:

$$F_t = S_t e^{(r+c)t}$$

where r is the risk-free rate and c is the storage or carrying cost. (3 marks)

- (b) Compare the theoretical futures price with the observed market price. Identify periods of potential *cash-and-carry arbitrage*. (2 marks)
- (c) Identify periods where *reverse cash-and-carry arbitrage* may exist. (2 marks)
- (d) Using three futures contracts with different maturities, construct a *butterfly spread* and analyse its profitability. (3 marks)

Question 6: Climate and Freight Derivatives Analysis (10 Marks)

Choose one dataset related to climate or shipping markets.

Possible variables include:

- Temperature
- Rainfall
- Frost days

- Baltic Dry Index (freight rates)
 - Carbon futures prices
- (a) Construct a historical time series dataset (at least 10 years if available). Plot the series and discuss trends and seasonality. (3 marks)
- (b) Estimate a statistical model (e.g. ARIMA or seasonal model) to forecast the variable. (3 marks)
- (c) Construct a derivative index such as:
- Heating Degree Days (HDD)
 - Cooling Degree Days (CDD)
 - Freight rate index
- (2 marks)
- (d) Design a derivative contract based on the index and simulate its payoff under different scenarios. (2 marks)

Question 7: Destabilisation Effect of Futures Markets on Spot Prices (10 Marks)

The introduction of futures markets is sometimes argued to increase spot market volatility through speculative trading. Using the *spot and futures dataset* from Question 2, examine whether futures trading destabilises the spot market.

- (a) Compute daily return volatility for the spot price using a rolling window (e.g., 30-day volatility). Plot the volatility series. (2 marks)
- (b) Estimate a $GARCH(1,1)$ model for the spot price returns:

$$r_t = \mu + \epsilon_t$$

$$\epsilon_t \sim N(0, h_t)$$

$$h_t = \omega + \alpha \epsilon_{t-1}^2 + \beta h_{t-1}$$

Interpret the persistence parameters. (3 marks)

- (c) Estimate a *GARCH model including futures market information*, for example by including futures return or trading volume as an explanatory variable in the variance equation. (2 marks)
- (d) Compare volatility before and after major futures trading activity (or across high vs low futures trading periods). Discuss whether the evidence supports the *destabilisation hypothesis* or the *market stabilisation hypothesis*. (2 marks)

Expected Deliverables

The final report should include:

- Data description
- Econometric methodology
- Estimation results
- Graphs and tables
- Economic interpretation
- Code appendix